
Design Integration Solutions

Given a use case, identify the integration pattern that meets business requirements.

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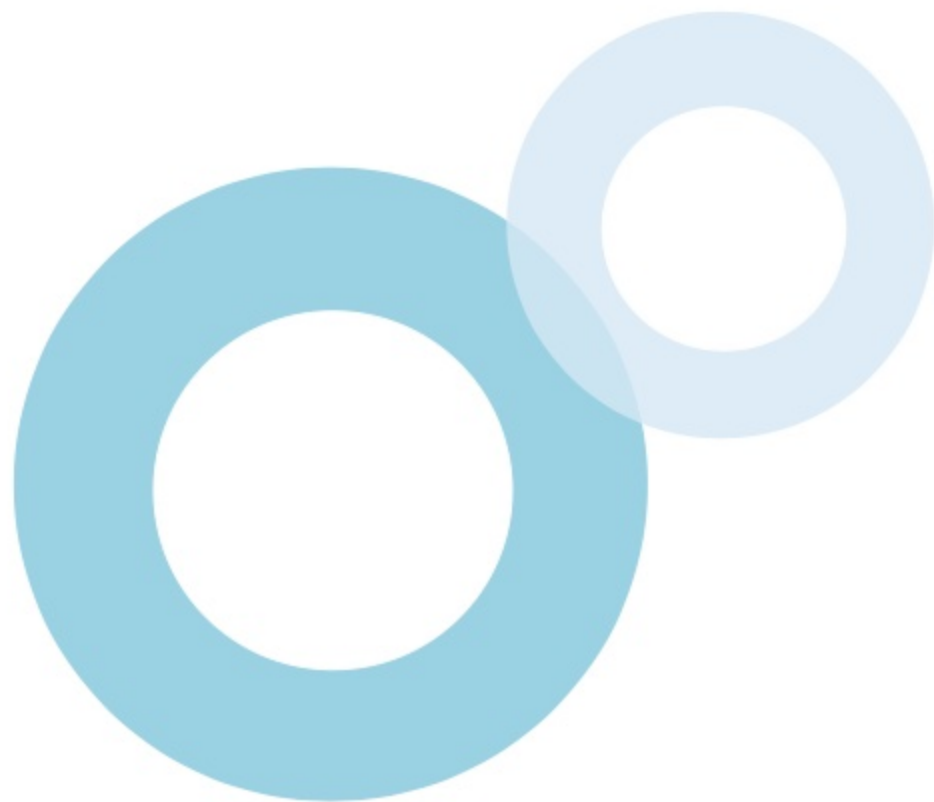


Requirements & Solutions



After studying this topic, you should be able to:

-  Define integration patterns and describe how they are used to integrate systems.
-  Describe the use of various integration patterns for integrating Salesforce with external systems.
-  Given a business scenario, explain how integration patterns can be utilized to meet a company's integration requirements.



Introduction

This topic describes various **integration patterns** and their **use cases** for designing integrations between Salesforce and remote systems. An integration pattern describes the design and approach for integrating the Salesforce platform with external systems. Integration can focus on **process**, **data**, or **virtual integration**.

The integration pattern called **Remote Process Invocation - Request and Reply** is used when Salesforce needs to perform a synchronous callout to a remote system and wait for a reply. **Remote Process Invocation - Fire and Forget** is used to send information to a remote system when an immediate response is not required. **Batch Data Synchronization** is used for exporting, transforming, and loading data between Salesforce and a remote system. **Remote Call-In** is used when a remote system needs to perform an API call to Salesforce to retrieve, create, update, or delete data. **Data Virtualization** is used to give Salesforce users access to data stored in a remote system. **UI Update Based on Data Changes** focuses on the use of Streaming API to update Salesforce UI whenever an event occurs. The **Publish/Subscribe** architecture involves the use of platform events, which can be used by Salesforce and a remote system to communicate events.



Integration Patterns

Remote Process Invocation Request & Reply

This integration pattern is used when Salesforce needs to perform a synchronous callout to a remote system and wait for a reply.



Remote Process Invocation Fire & Forget

This integration pattern is used when Salesforce needs to perform an asynchronous callout to a remote system and not wait for an immediate response.



Batch Data Synchronization

Batch Data Synchronization is used to synchronize data between Salesforce and a remote system by exporting, transforming, and importing data.



Remote Call-In

A remote system can perform a callout to Salesforce to retrieve, create, update, or delete data. This pattern can utilize SOAP, REST, or Bulk API.



Data Virtualization

External data can be virtualized or exposed in Salesforce using Salesforce Connect.



UI Update Based on Data Changes

Streaming API can be used to publish notifications for real-time UI updates when Salesforce events occur.



Integration Patterns

Integration patterns or strategies are used to integrate Salesforce with **third-party applications**.

❖ DESIGN & APPROACH

An integration pattern describes the design and approach for integrating **Salesforce** with external systems or applications.

❖ INTERACTIONS

Integration patterns can be used to identify how **systems**, including their **components** and **services**, interact in an integration solution design.

❖ INTEGRATION INITIATIVES

The integration patterns are classified into the following categories: **process integration**, **data integration**, and **virtual integration**.



Integration Initiatives

There are three types of integration initiatives.

❖ PROCESS INTEGRATION

Process integration focuses on **extending business processes and services across systems**. It typically utilizes events that trigger activity in one system or facilitate transactions between two systems.

❖ DATA INTEGRATION

This focuses on **integrating and/or synchronizing data between two or more systems**. It includes aspects such as data integrity, data governance, data flow design, and data migration.

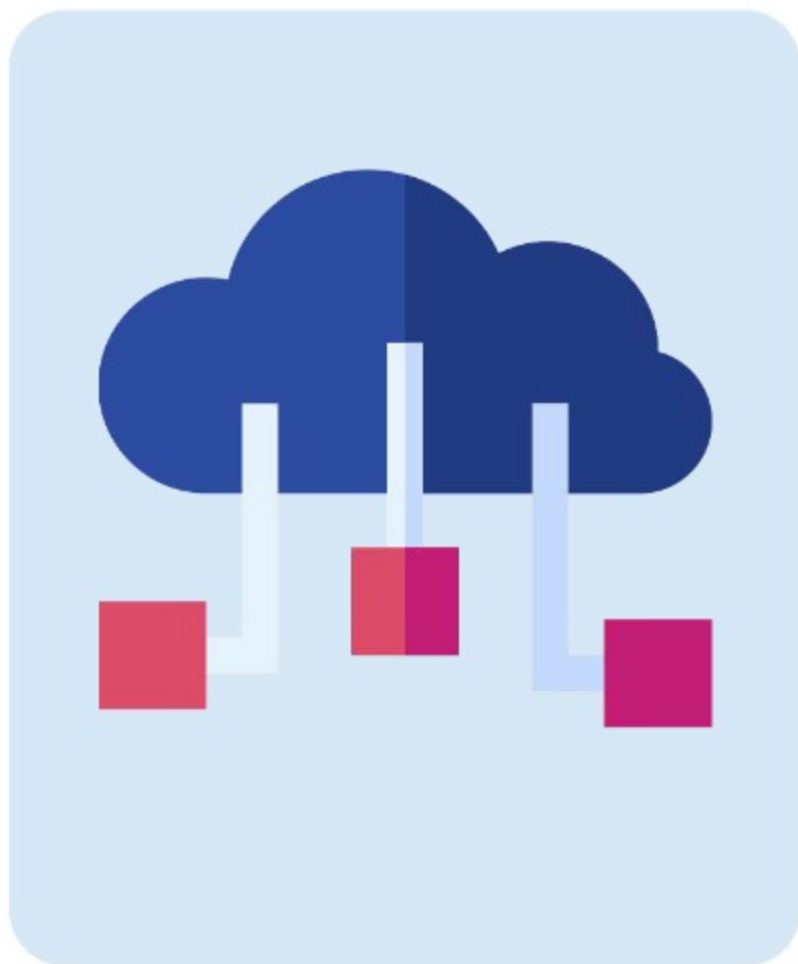
❖ VIRTUAL INTEGRATION

This type of integration addresses the need to enable users to interact with data that resides in an **external source**. Data is accessed in **real time**, which avoids the need for data replication across systems.



Integration Patterns

The following are the various integration patterns supported by the Salesforce Lightning platform:



INTEGRATION PATTERNS

- ❖ Remote Process Invocation - Request and Reply
- ❖ Remote Process Invocation - Fire and Forget
- ❖ Batch Data Synchronization
- ❖ Remote Call-In
- ❖ Data Virtualization
- ❖ UI Update Based on Data Changes

Remote Process Invocation - Request and Reply

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Remote Process Invocation - Request and Reply

In this integration pattern, Salesforce calls a remote system **synchronously** and waits for a reply.

❖ RESPONSE

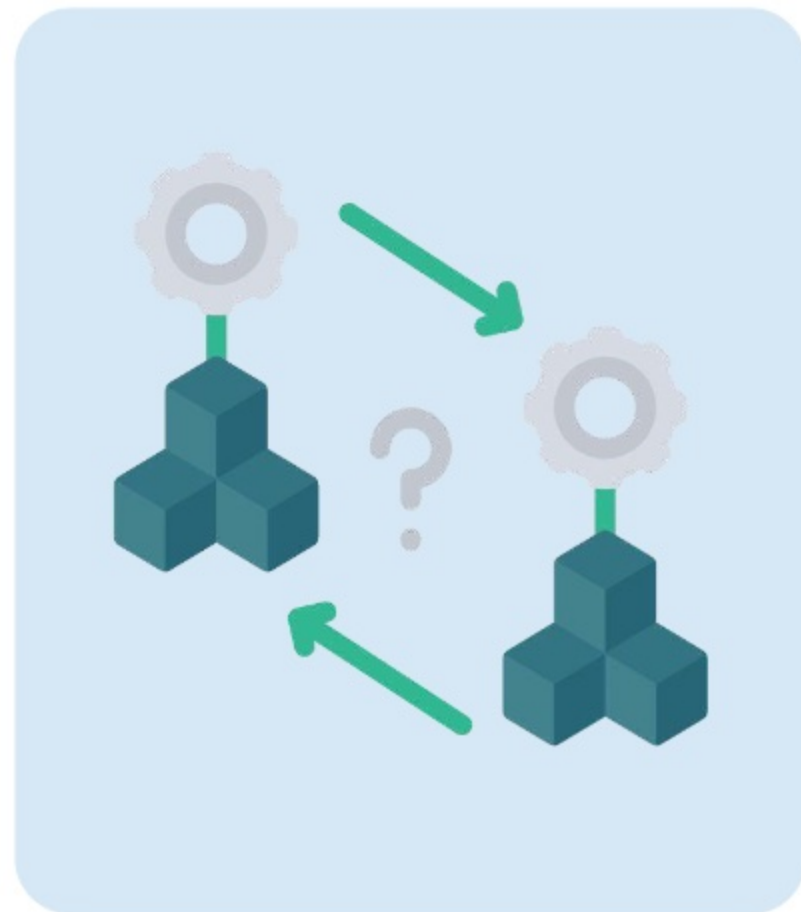
Salesforce processes the **response** from the remote system within the **same transaction** as the **initial call**.

❖ VISUALFORCE

A custom Visualforce page or button can be used to **initiate the callout** to the remote system **synchronously**.

❖ CALLOUT

A trigger that is invoked from Salesforce data **change events** or a batch **Apex job** can also be used for a callout.



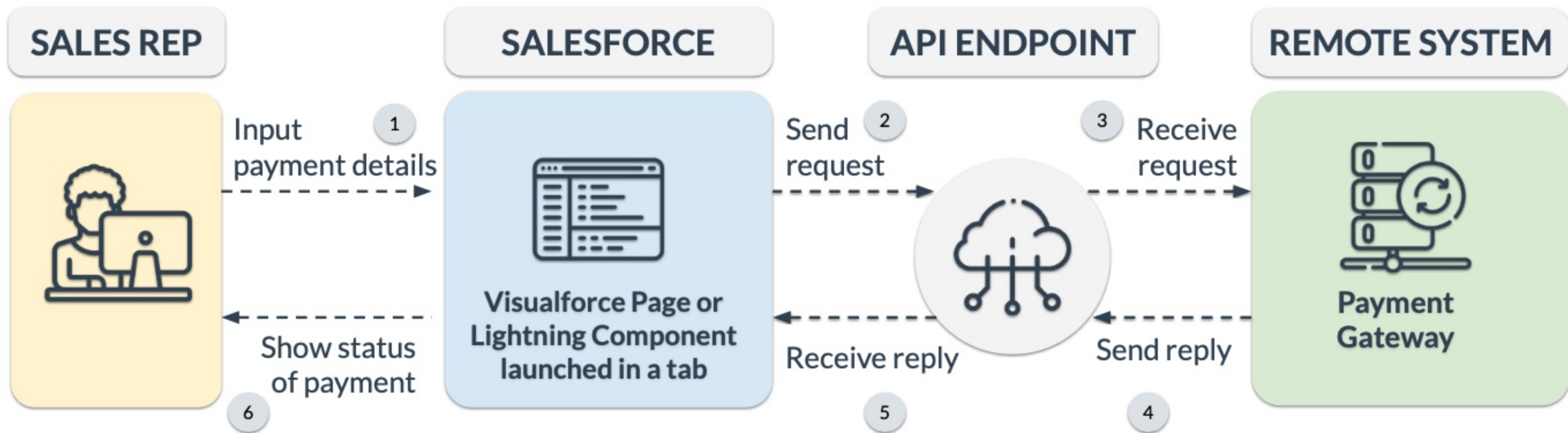
Visualforce Page

The following **high-level steps** describe the process when a Visualforce page is used in the **Remote Process Invocation - Request and Reply** integration pattern:



Request and Reply Example Using a Visualforce Page

A sales representative creates an **order** in Salesforce, and an **external system** processes the **order payment**. Once the payment has been processed, Salesforce updates the **order status**.



Request and Reply Solutions

The table below lists the different types of integration solutions that utilize the **Request and Reply** integration pattern.

SOLUTION	SUITABILITY
External Services	Best
Synchronous SOAP or REST Callout from a Lightning Component	Best
Synchronous SOAP or REST Callout from a Visualforce Page or Button	Best
Synchronous SOAP or REST Callout from a batch Apex job	Suboptimal

Remote Process Invocation - Fire and Forget

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Remote Process Invocation - Fire and Forget

Salesforce calls a **remote system** to initiate a process and sends the required information to that process, but **does not wait for a response** from the remote system.

❖ SALESFORCE UPDATE

The remote system can optionally update Salesforce in a **separate transaction**.

❖ OUTBOUND MESSAGE

- **Flow-driven outbound messaging** is the recommended solution for this type of integration pattern.
- It can be used to send information asynchronously to a specific **endpoint URL** of a remote system.



Remote Process Invocation - Fire and Forget

❖ ASYNCHRONOUS CALLOUT

Custom **Visualforce pages**, **triggers** invoked from Salesforce data **change events**, or **batch Apex jobs** can also be used for an asynchronous callout to the remote system.

❖ ASYNCHRONOUS PROCESS

After receiving a message, the remote system responds and indicates receipt of the message, and then **asynchronously processes the message**.

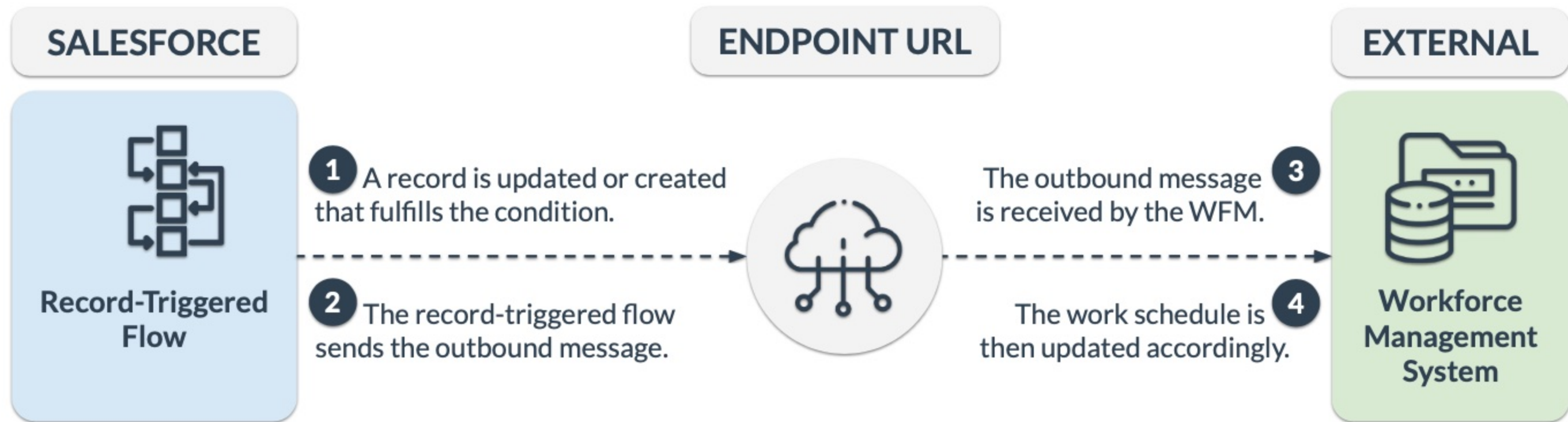
❖ AFTER RECEIVING REQUEST

The remote system hands control back to Salesforce before it begins to process the message, which means that Salesforce **does not have to wait** for the processing to complete.



Fire and Forget Example Using an Outbound Message

This integration pattern can be illustrated by a flow that uses an **outbound message action** to send a message to an **external system** to update details regarding an agent's work schedule.



Fire and Forget Solutions

The table below lists the different types of integration solutions that utilize the **Fire and Forget** integration pattern.

SOLUTION	SUITABILITY
Process-Driven Platform Events	Best
Customization-Driven Platform Events	Good
Workflow-Driven Outbound Messaging (with or without callback)	Good
Asynchronous SOAP or REST Callout from a Lightning Component or Visualforce Page	Suboptimal
Asynchronous SOAP or REST Callout from a Synchronous Apex Trigger	Suboptimal
Asynchronous SOAP or REST Callout from a Batch Apex job	Suboptimal

Batch Data Synchronization

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Batch Data Synchronization

This integration pattern focuses on **synchronizing data** between systems, either by **importing data** from an external system, or by **exporting data** into an external system.

❖ PERIODIC BATCH PROCESSING

Data can be exported from a source system, reflecting changes over a certain period, such as **daily or weekly**. **Large amounts of data** can be **exported** or **imported**, and the records are processed in **batches**.

❖ ETL TOOL

A **third-party ETL** (Extract, Transform, and Load) tool can be used for a one-time or ongoing data export or import.

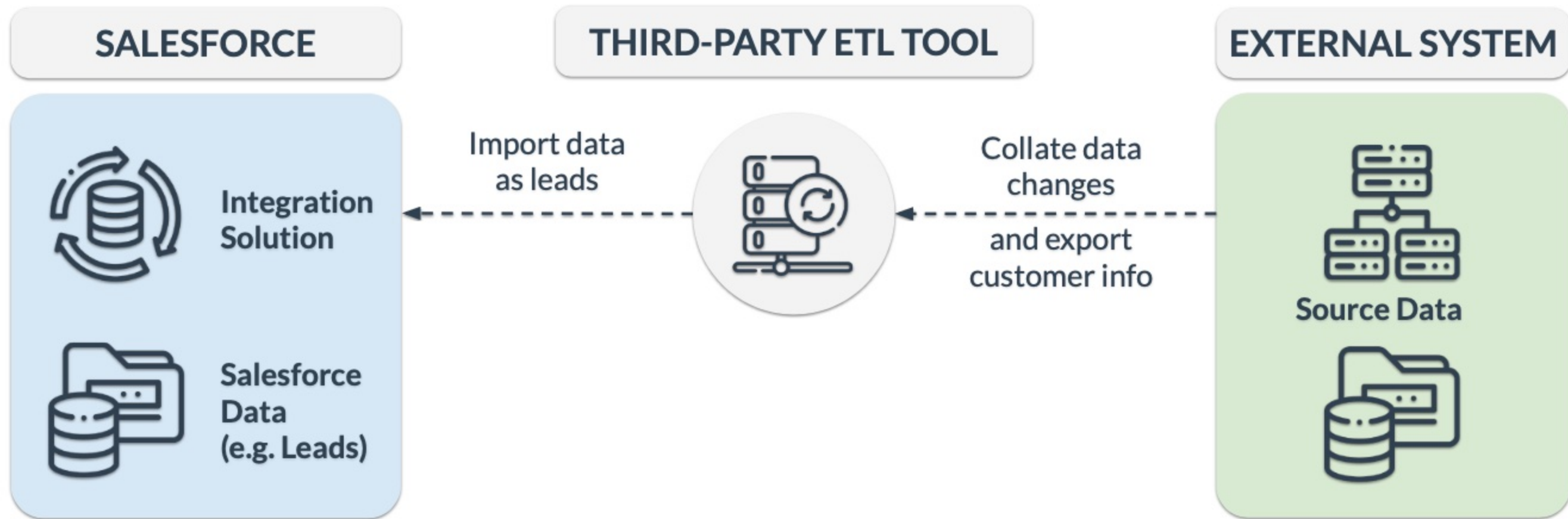
❖ END-USER OPERATIONS

This integration pattern ensures that imports and exports **do not interfere** with end-user operations during business hours, and is particularly suitable for large amounts of data.



Batch Data Synchronization Example

Below is an example of an integration that uses **Batch Data Synchronization**.



Batch Data Synchronization Solutions

The table below lists the different types of integration solutions that can be utilized in **batch data synchronization** scenarios.

SOLUTION	SOURCE	TARGET	SUITABILITY
Salesforce Change Data Capture	Salesforce	Remote System	Best
Data Replication Using an ETL Tool	Remote System	Salesforce	Best
Data Replication Using an ETL Tool	Salesforce	Remote System	Good
Remote Call-In	Remote System	Salesforce	Suboptimal
Remote Process Invocation	Salesforce	Remote System	Suboptimal

Remote Call-In

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Remote Call-In

A **remote system** can create, retrieve, update, or delete **data stored in Salesforce**.

❖ SOAP OR REST API

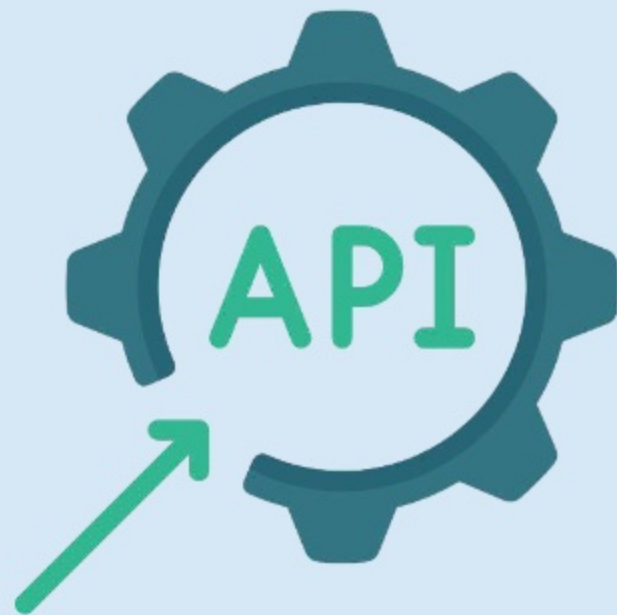
SOAP or REST API can be used by remote systems to **query, create, update, and delete data, obtain metadata** about an organization, and **run utilities** to perform administrative tasks.

❖ BULK API

Using Bulk API is an optimal solution for loading **large sets of data**.

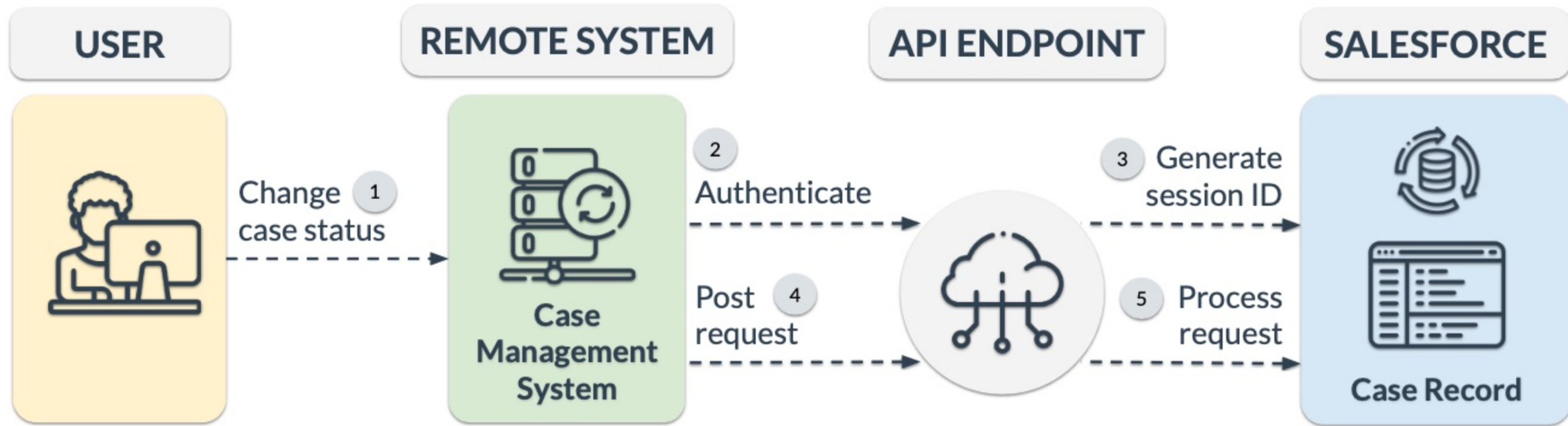
❖ APEX WEB SERVICES

Apex Web service or **Apex REST service** can also be used to implement this integration pattern, but they are considered **suboptimal solutions** since they are typically used **only when** full transactional support or custom logic is required.



Remote Call-In Example

This integration pattern can be used to **update a case record in Salesforce** once its status has been updated in a remote system (external website) using REST API.



Remote Call-In Solutions

The table below lists the different types of integration solutions that utilize the **Remote Call-In** integration pattern.

SOLUTION	SUITABILITY
SOAP API	Best
REST API	Best
Apex Web Service	Suboptimal
Apex REST Service	Suboptimal
Bulk API 2.0	Optimal (for large data volume operations)

Data Virtualization

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Data Virtualization

In **Data Virtualization**, Salesforce can access external data in real time. While the **external system contains and processes the data**, Salesforce users can view and update it in real-time.

❖ DATA IS NOT STORED

Data from the external system is not stored in **Salesforce**.

❖ SALESFORCE CONNECT

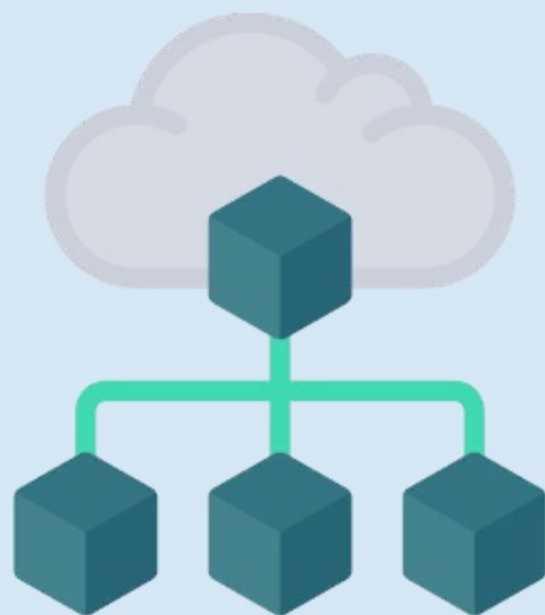
Salesforce Connect can be used to expose the external data and allow users to view, search, and modify records in an external system.

❖ EXTERNAL DATA SOURCE & OBJECTS

Salesforce Connect can be used to create an **external data source** and **external objects** for the records in the external system.

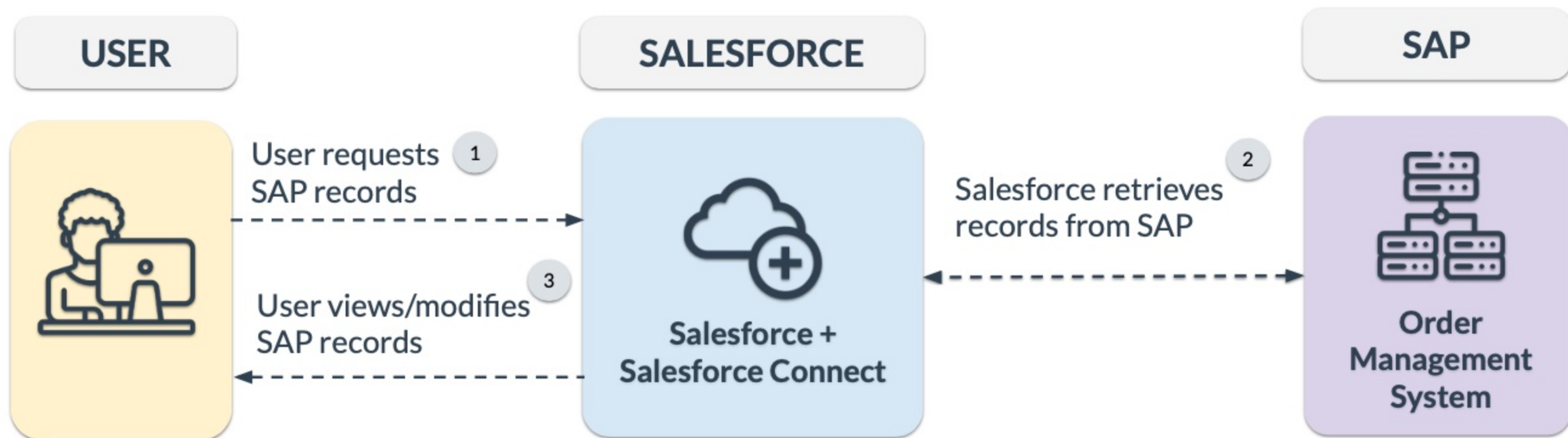
❖ WRITABLE EXTERNAL OBJECTS

Writable external objects can be enabled in an external data source to **allow modification of external records**.



Data Virtualization Example

Sales orders from **SAP** can be accessed in real time and modified directly in Salesforce by using **Salesforce Connect**.



Data Virtualization Solutions

The table below lists the different types of integration solutions that utilize the **Data Virtualization** integration pattern.

SOLUTION	SUITABILITY
Salesforce Connect	Best
Synchronous REST or SOAP Callout Using a Lightning Component, Visualforce Page, or Button	Suboptimal

UI Update Based on Data Changes

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UI Update Based on Data Changes

The **user interface** can be updated in **real-time** whenever an event occurs in Salesforce.

❖ STREAMING API

Salesforce Streaming API is the recommended solution to provide real-time UI updates triggered by **Salesforce events**.

❖ CHANGE EVENTS

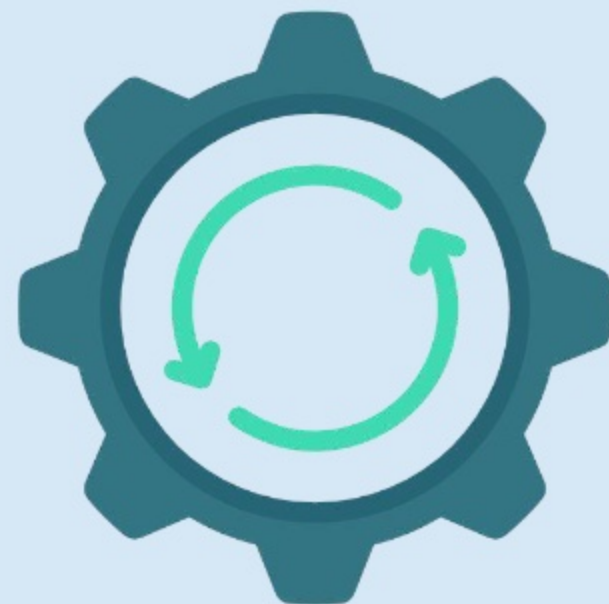
A platform event or PushTopic can **send a notification** to certain users based on data change events in Salesforce.

❖ CUSTOM UI

A custom Lightning component or Visualforce page can be created to make the notifications **visible** to the users.

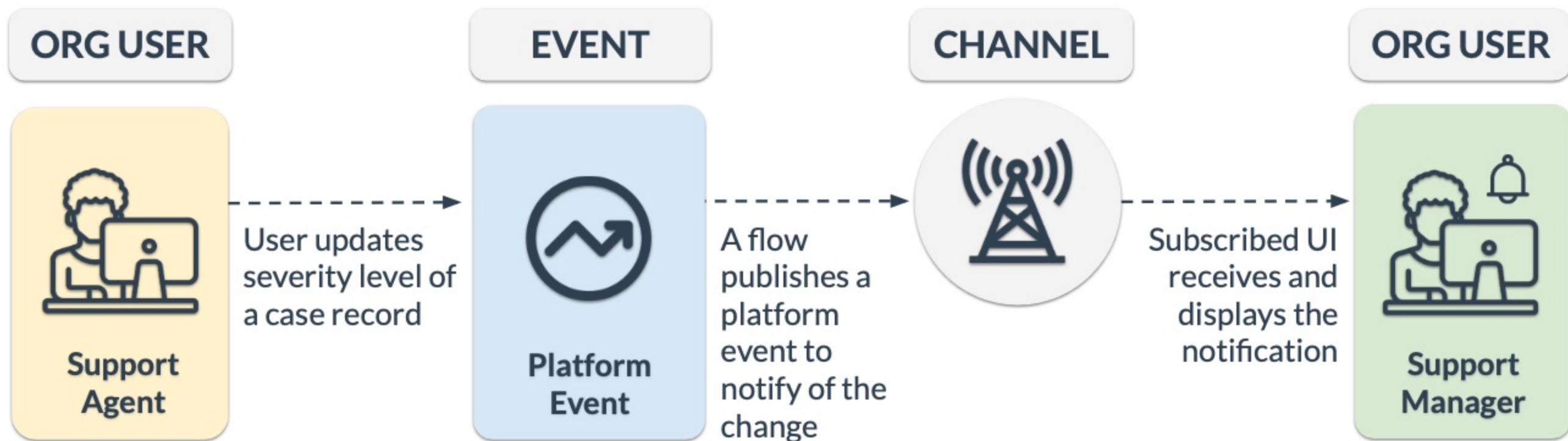
❖ BENEFIT

This allows Salesforce UI updates **without refreshing the screen** and potentially losing unsaved work.



UI Update Based on Data Changes Example

This integration pattern can be used to allow Salesforce users to **receive notifications** without reloading the UI when **a user updates a record**.



Publish/Subscribe

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Publish/Subscribe

The **Publish/Subscribe** architecture involves the use **Platform Events** to deliver real-time notifications in **Salesforce** or an **external system**.

❖ EVENT DRIVEN ARCHITECTURE

Platform Events make use of **event-driven architecture**, which consists of **event producer**, **event consumer**, and **channel**.

❖ EVENT PRODUCER

The event producer **publishes an event message** over a channel when an **event** occurs. For example, it can be a record update in Salesforce.

❖ CHANNEL

The channel (or **event bus**) is the **conduit** that **transmits** the message.

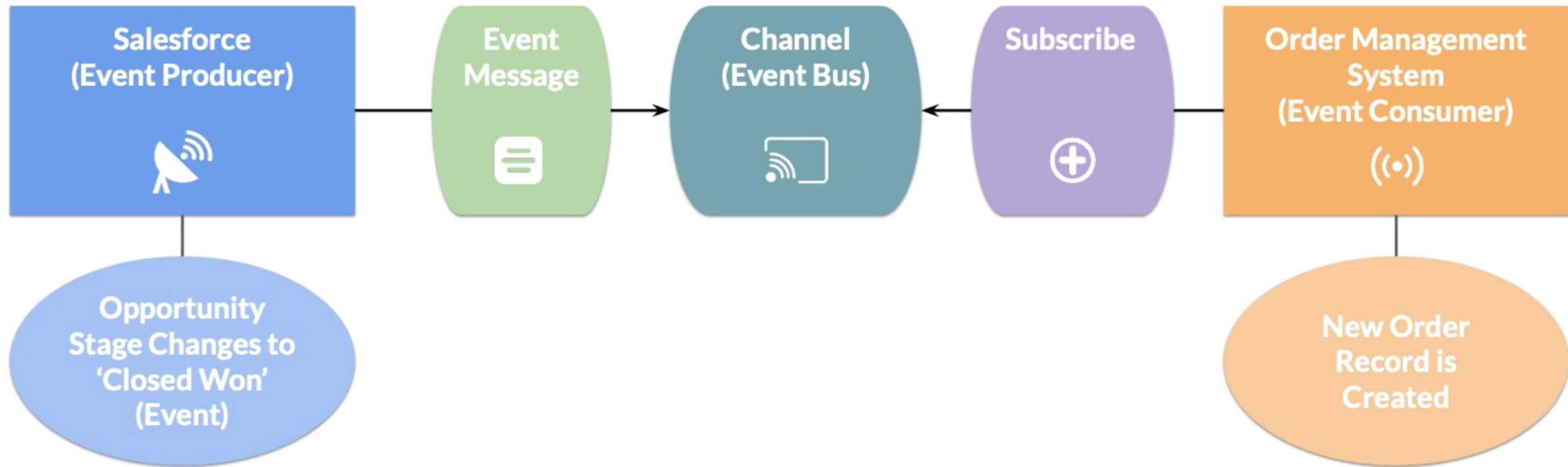
❖ EVENT CONSUMER

An event consumer or subscriber **subscribes** to the channel and **receives** the event message. For example, an external user can receive a notification when there is a record update in Salesforce.



Publish/Subscribe Example

Salesforce can **publish** a **platform event message** when an opportunity is closed. An order management system can **subscribe** to the **channel** and **receive** the message to create an order.



Business Scenario

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Business Scenario: Part 1

Read the business scenario below and determine the most appropriate solutions for the company's requirements.



Cosmic Electronics is a multinational company that sells various types of electronic products. It has offices in the United States and several other countries like the United Kingdom, France, and Italy. It currently employs more than 50,000 people worldwide and has sold its products to more than 50 million customers, comprising both individuals and businesses.

It uses multiple enterprise systems to manage organizational and customer data. Salesforce is primarily used for managing accounts, contacts, leads, opportunities, and cases. An enterprise resource planning (ERP) system is used for accounting, order management, and supply chain management. A workforce management (WFM) system is used for human resource management, performance management, and training delivery.

Business Scenario: Part 2

Read the business scenario below and determine the most appropriate solutions for the company's requirements.



However, none of the enterprise systems are currently integrated, and each system is used separately by internal users to perform tasks. For example, when an opportunity is closed in Salesforce, a sales representative needs to manually create an order in the ERP system, which goes through various stages until the customer's order is delivered.

The Chief Technology Officer would like to make the company's business processes more efficient by integrating the systems based on the internal users' requirements. The company has hired a Salesforce Architect who needs to analyze several requirements for an integration project and design solutions to meet them. She needs to identify integration patterns while recommending suitable solutions.

Requirements & Solutions

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Requirement 1



SCENARIO

The sales representatives of the company use a Lightning web component in Salesforce to manage opportunities. Once a sales representative marks an opportunity as 'Closed Won', they need to access the ERP system to create a sales order for the customer by entering all the essential details, such as the products and shipping address. The two systems need to be integrated such that a sales order is created automatically in the external system and the sales representative immediately gets a response from the system in real time, confirming the successful creation of the sales order for the customer.



Solution 1

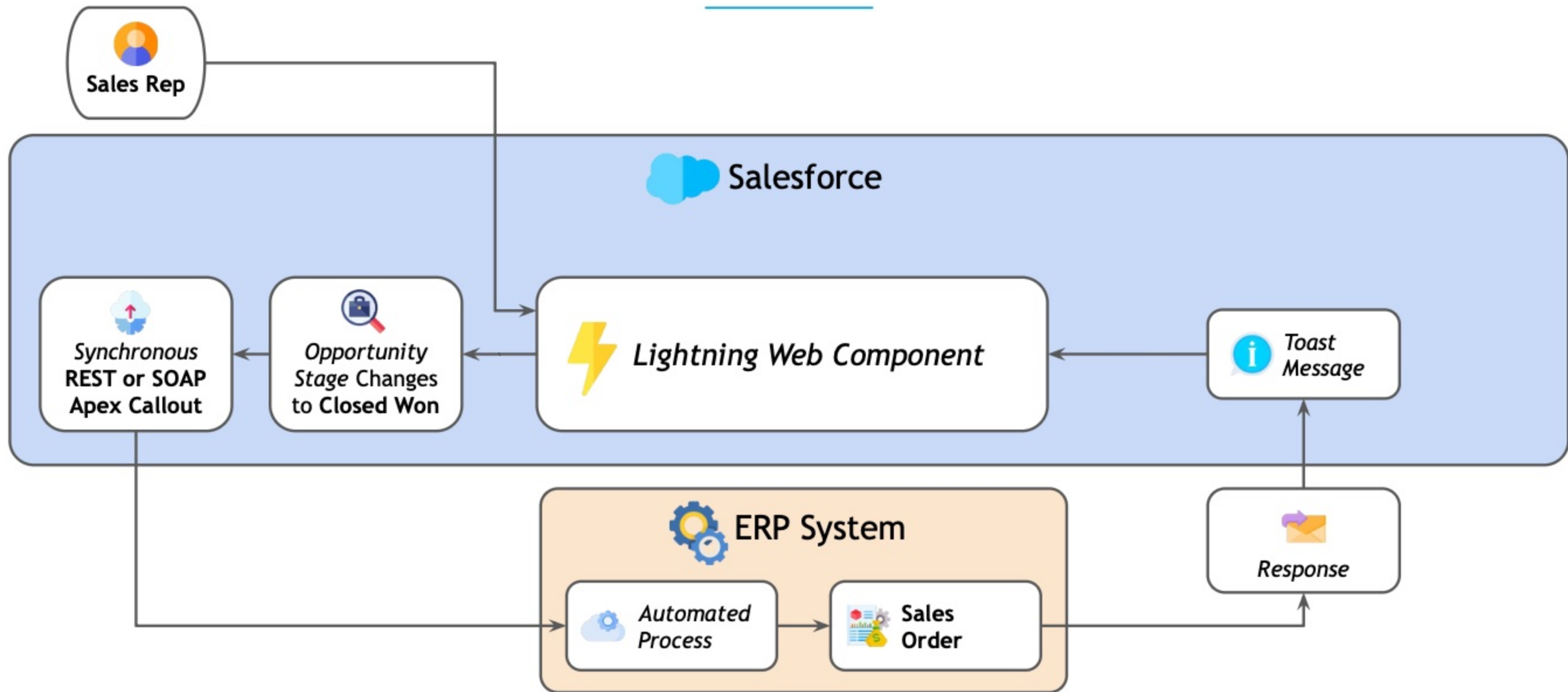


SOLUTION

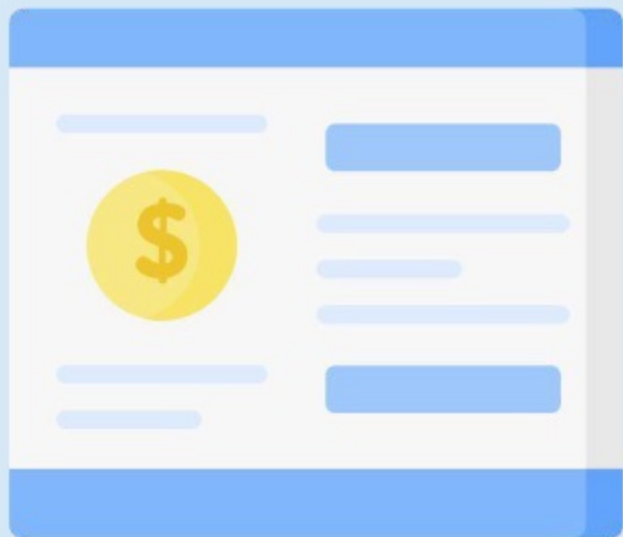
Salesforce and the ERP system can be integrated using the integration pattern called '**Remote Process Invocation - Request and Reply**'. When the stage of an opportunity is changed to 'Closed Won', the Lightning web component can use Apex code to invoke a **REST or SOAP callout** to the ERP system **synchronously** and **wait for a reply**. The callout can contain details about the sales order. The ERP system can use the details to create a new sales order automatically, and then immediately send a response to Salesforce in real time to provide confirmation. A toast message can be displayed in the Lightning web component to let the sales representative know that the sales order has been created in the ERP system.



Solution 1



Scenario 2

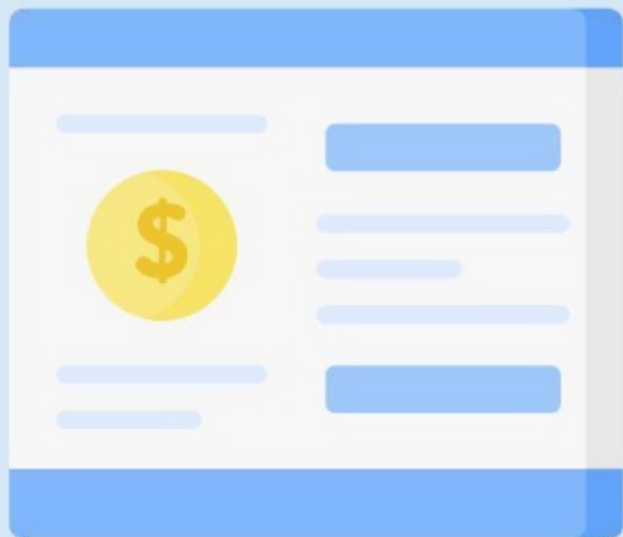


SCENARIO

Sales associates often need to manually update account information in Salesforce and the ERP system. For instance, when a company places an order for a large number of products, a sales associate changes the rating of that company's account to 'Hot' both in their Salesforce org and ERP system.

The sales director has requested that whenever there is an account update in Salesforce, the same update should be automatically applied to the same account record in the ERP system, which is equipped with SOAP web services. The sales associate who performs the update in Salesforce does not need to wait for a response from the ERP system.

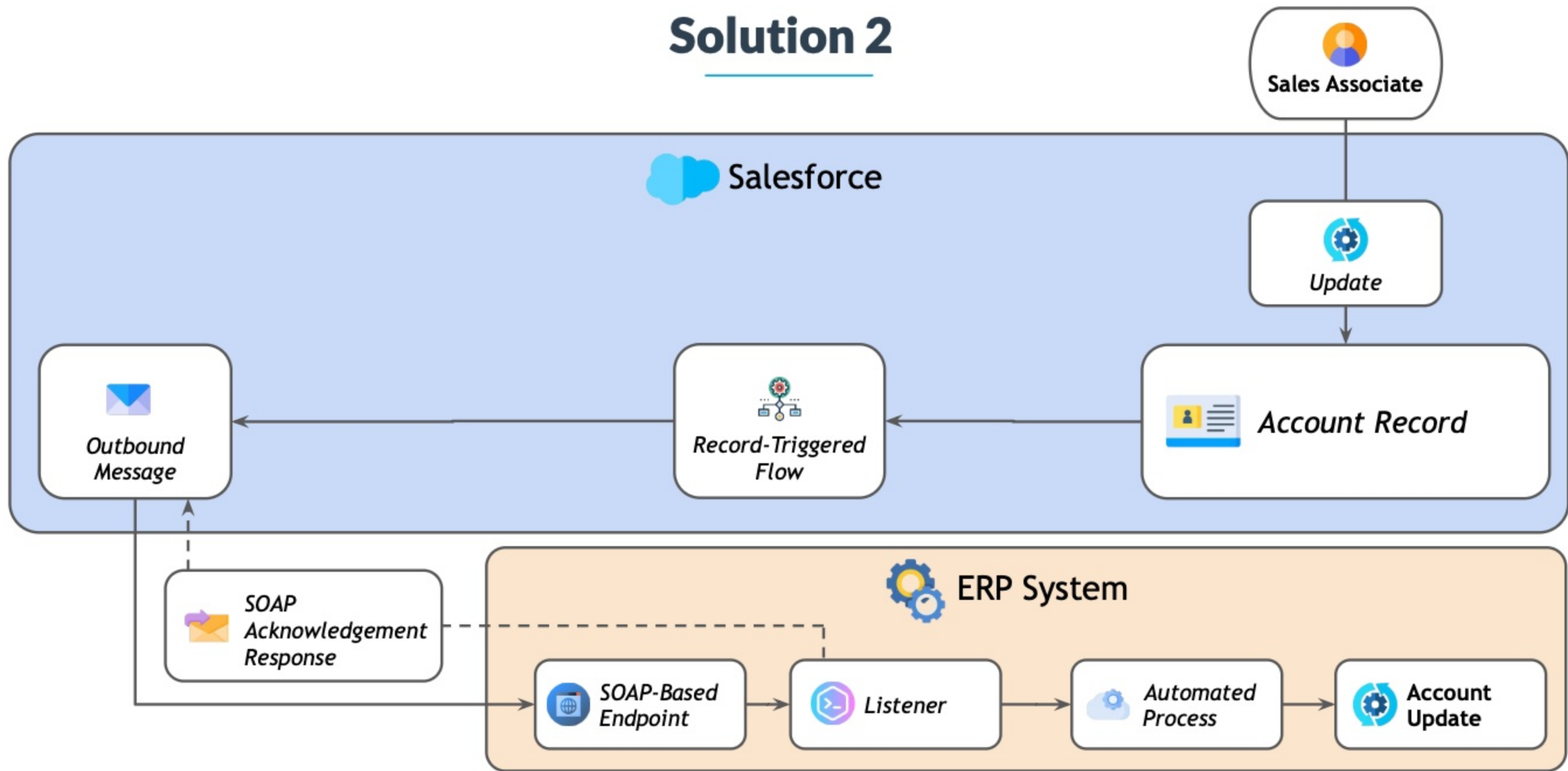
Solution 2



SOLUTION

Salesforce and the ERP system can be integrated using the integration pattern called '**Remote Process Invocation - Fire and Forget**'. When an account record is updated in Salesforce, a record-triggered flow can asynchronously send an **outbound message** containing the Account fields to a **SOAP-based endpoint** of the ERP system. Once the listener in the ERP system receives the outbound message, it can return an acknowledgment response to Salesforce to indicate the successful delivery of the message. The ERP can then process the data contained in the message. The sales associate who updates the account record in Salesforce can save the record **without waiting for a response**.

Solution 2



Requirement 3



SCENARIO

The workforce management (WFM) system contains data about hundreds of thousands of events that were scheduled by the company's employees during the current fiscal year. The sales director would like to add this data to Salesforce for reference. Any events that are scheduled in the future should be added to Salesforce at the end of the day.



Solution 3

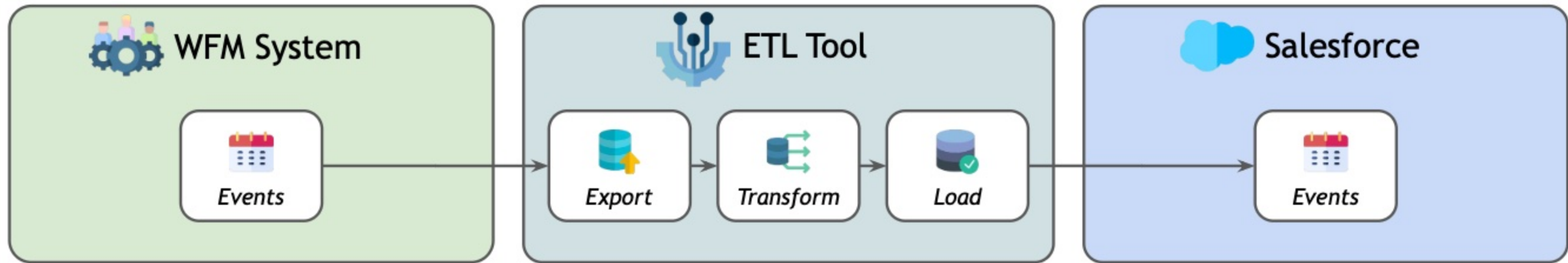


SOLUTION

'**Batch Data Synchronization**' can be utilized to add events from the workforce management (WFM) system to Salesforce **initially and periodically**. The existing events scheduled during the current fiscal year can be exported from the WFM system, transformed based on the target data model, and then loaded into Salesforce. A suitable **ETL (Extract, Transform, Load)** tool can be utilized for it. Any future events can be exported and imported into Salesforce automatically at the end of the day using the same tool.



Solution 3



Scenario 4



SCENARIO

The WFM system is the primary source of data about the company's employees, such as their mailing address, phone number, and email address. Whenever employee information needs to be updated, an HR specialist performs the update in the WFM system. However, if an employee can access Salesforce, the specialist also needs to manually update the employee's user record in Salesforce. The HR director would like to automate this process.

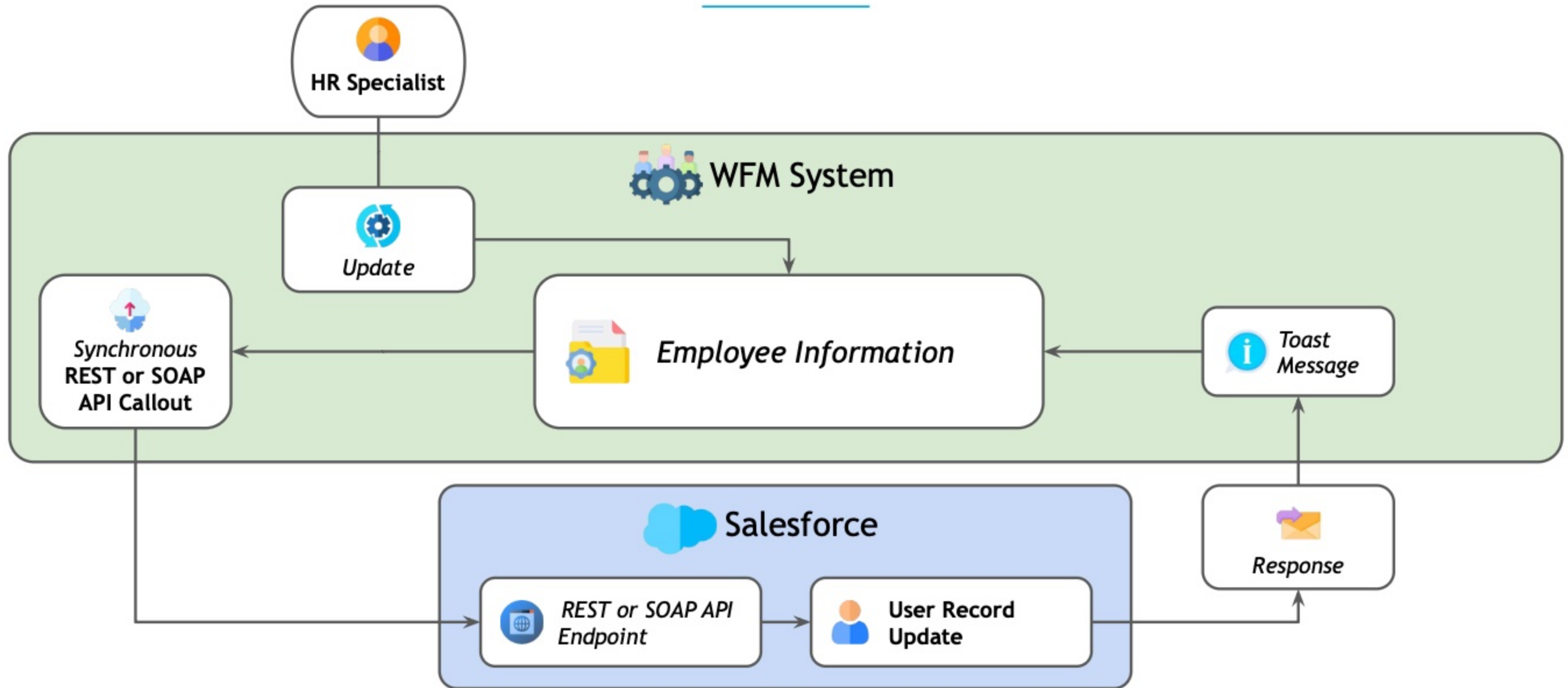
Solution 4



SOLUTION

This requirement can be met by integrating the two systems using the '**Remote Call-In**' integration pattern. The WFM system can use **REST API** or **SOAP API** to perform a **synchronous** callout to Salesforce that automatically updates the user record of the employee whenever the corresponding information is updated in the WFM system.

Solution 4

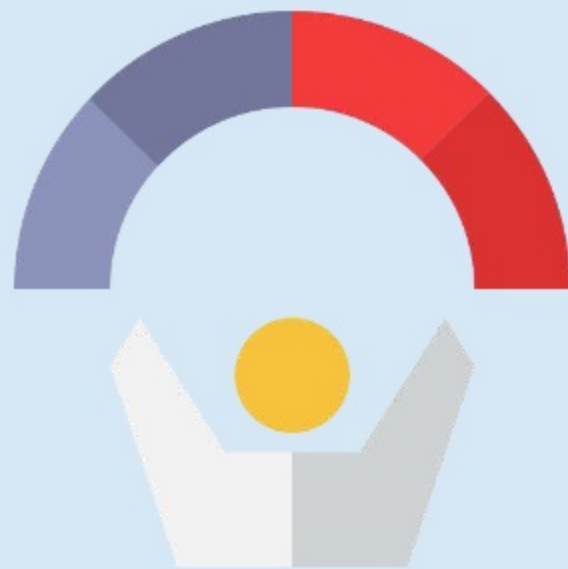


Requirement 5



SCENARIO

The workforce management (WFM) system contains millions of employee performance reviews. The members of the executive management can access them by logging in to the system. However, since they use Salesforce more frequently, they would like to gain quick access to the performance reviews in Salesforce. In addition, they need the ability to run reports on them and collaborate on them using the Chatter feed. However, due to the large volume of records, the Chief Technology Officer would prefer not loading them into Salesforce for this use case.

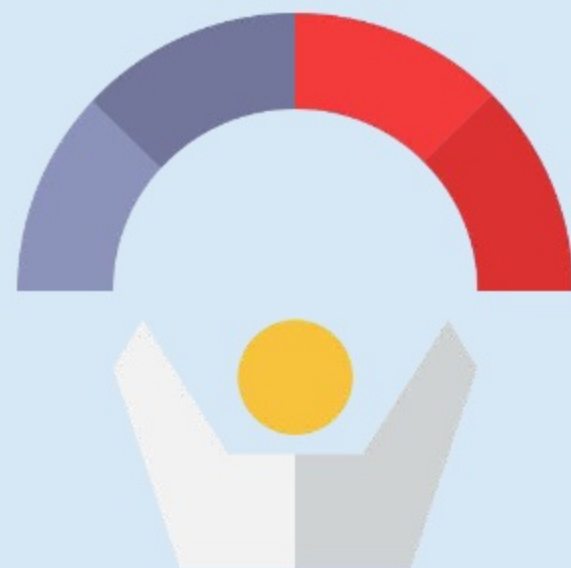


Solution 5

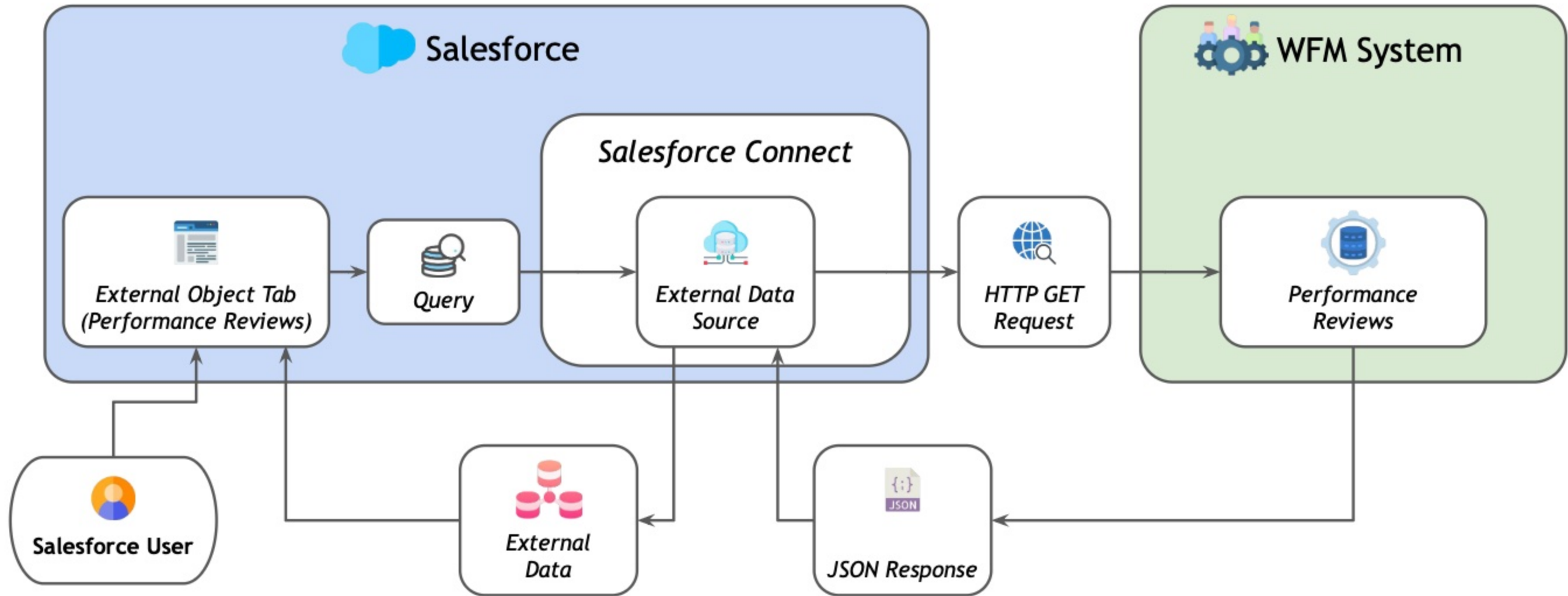


SOLUTION

This requirement can be met using '**Data Virtualization**'. **Salesforce Connect** can be set up to give Salesforce users real-time access to employee performance reviews stored in the WFM system. While the data is stored in the external system, Salesforce users can view or edit it. Furthermore, to allow collaboration on the external data, a Chatter feed can be enabled on the external object that represents the external data. It is also possible to run reports on the external data, but some features such as cross filters and bucket fields are not supported.



Solution 5



Scenario 6



SCENARIO

The support agents of the company frequently receive phone calls from customers who face payment issues. Once a payment issue has been resolved on the phone and the customer has made the payment, the support agent needs to check a third-party payment processing application to determine whether the payment was successful. The support director would like to automate this process. When the payment is successful, the support agent should see a real-time update in a Lightning web component that is used for case management.

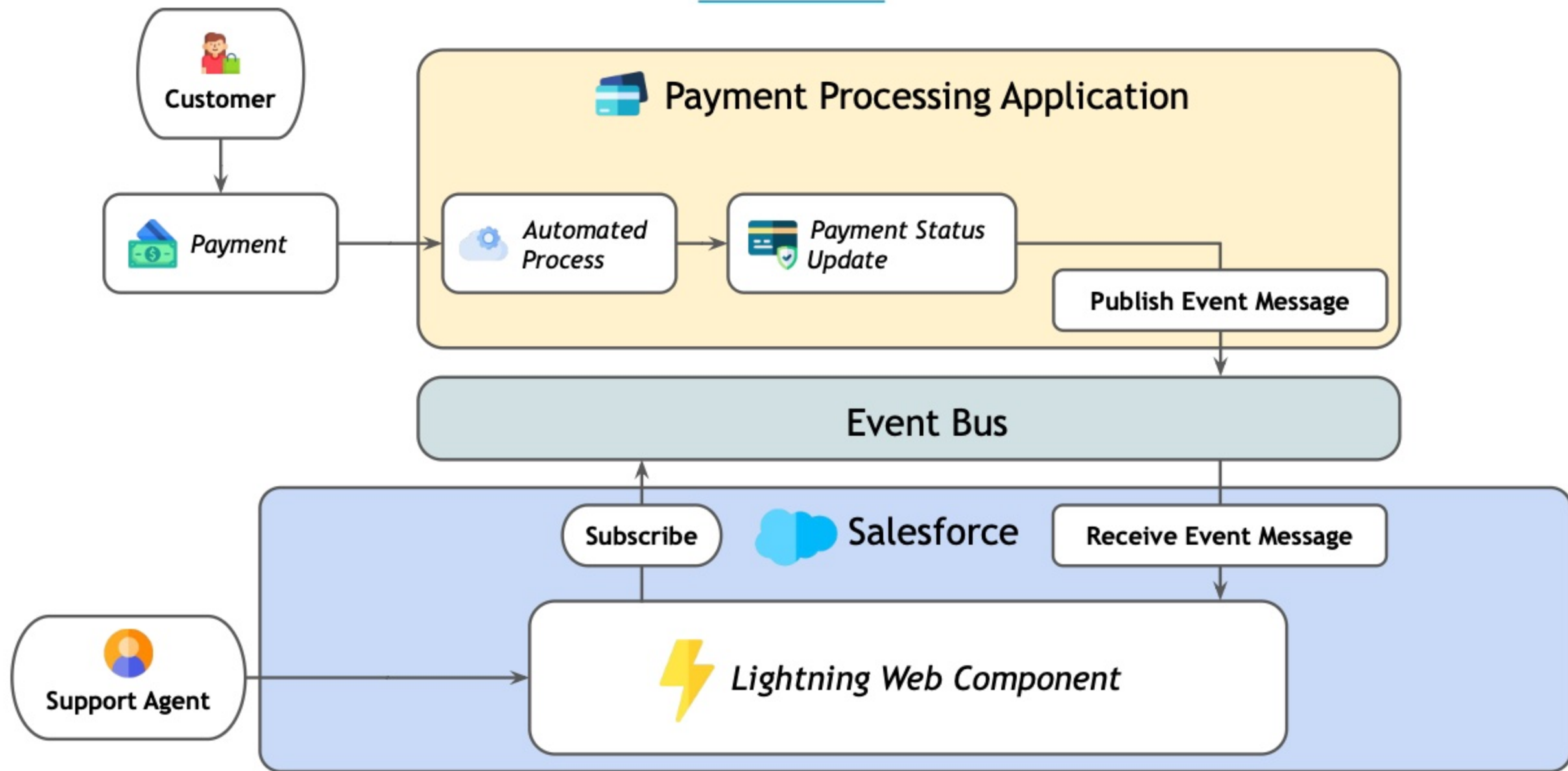
Solution 6



SOLUTION

This requirement can be met using the '**UI Update Based on Data Changes**' integration pattern. A **platform event** can be defined for payment status in Salesforce. The third-party payment processing application can publish a **platform event message** when a customer's payment is successful. The **Lightning web component** can subscribe to the **event bus/channel** to receive platform event messages. When an event message is received, a toast message can be displayed in the component to alert the support agent and let them know whether the payment has been successful.

Solution 6



Learn More

-  [Integration Patterns and Practices](#)
-  [Application Integration Patterns for Salesforce Lightning Platform](#)
-  [Change Data Capture Basics](#)
-  [Platform Event Basics](#)